

WASP-11b

R-1061

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_160115 · created 2026-07-30T11:45:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457402.65009 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1602 +/- 0.0079
Transit depth (Rp/Rs)^2	2.57 +/- 0.25 [%]
Orbital Inclination (inc)	89.88 +/- 0.36
Airmass coefficient 1 (a1)	0.9758 +/- 0.0085
Airmass coefficient 2 (a2)	0.02 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	#3 - [426, 277]
Optimal Aperture	3.34
Optimal Annulus	5.29
Transit Duration (day)	0.10974 +/- 0.00085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1602 $\pm$ 0.0079 vs 0.131 (deviation 3.7 σ)
Duration fit vs published	0.10974 d vs 0.1075 d (deviation 2.64 σ)
Mid-time O-C	3.6 min
Depth SNR	20.3
Residual scatter	0.87 %

Light curve

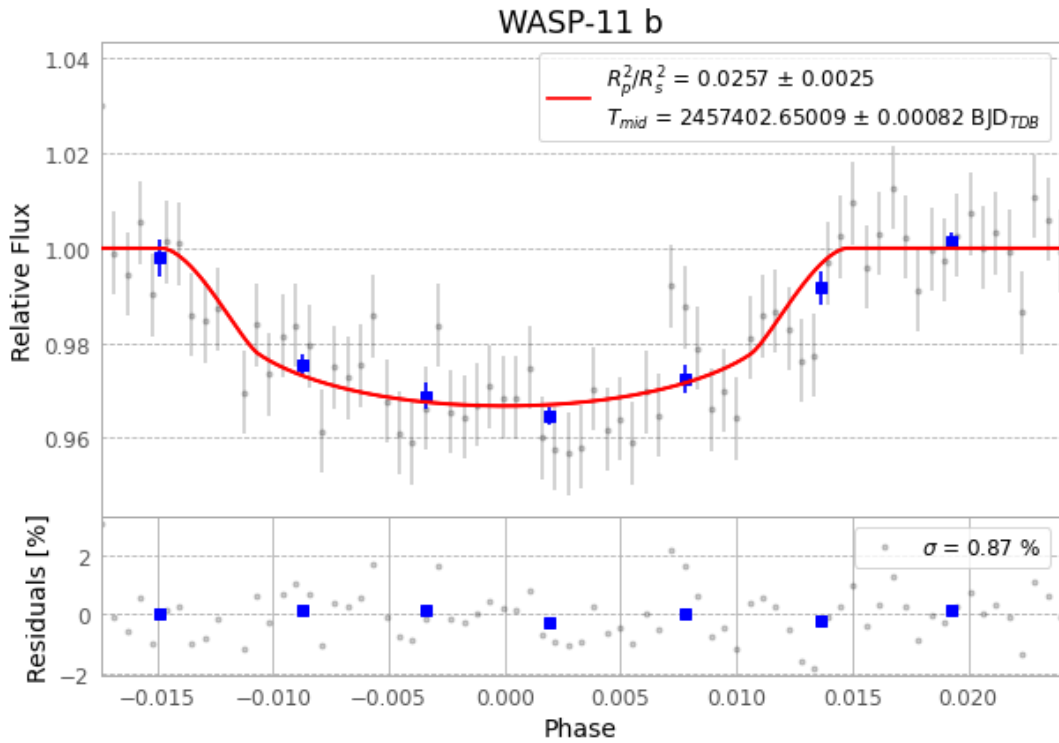


Figure 1 — FinalLightCurve_WASP-11 b_2016-01-14.png (SHA-256 3e8ba068ef666df4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a4119567761cfd2f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-10160115015712.FITS → HATP-10160115053913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-160115.log (d33af6706e35...), FinalLightCurve_WASP-11 b_2016-01-14.png (3e8ba068ef66...), FinalLightCurve_WASP-11 b_2016-01-14.csv (f235263c3dcd...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:45Z.